# INE Skill Check Lab Report
# Assessment Methodologies: Footprinting and Scanning CTF 1

**Target:** http://target.ine.local
**Assessment Type:** Footprinting and Scanning / Information Disclosure
**Tools Used:** Nmap, Curl, FTP, MySQL

## Executive Summary

This report documents the reconnaissance and enumeration performed against the target system target.ine.local. The objective of this assessment was to identify exposed services and discover hidden information within the environment. During the assessment multiple information disclosure issues were identified including exposed HTTP headers, hidden directories, anonymous FTP access, and database credentials stored in accessible files.

## Methodology

The assessment was performed using a structured reconnaissance methodology. First, service discovery was performed using Nmap to identify open ports and services. Next, HTTP enumeration was conducted using curl to inspect server responses and headers. Hidden directories were identified through manual enumeration. FTP services were accessed using anonymous login to retrieve exposed files, and database credentials were extracted and used to enumerate MySQL databases.

## Flags Captured

| Flag | Value |
|---|---|
| Flag 1 | FLAG1_afa965c7836048cf955a2181f0400af2 |
| Flag 2 | FLAG2_6947ca5b4e447f982d12226b459682e |
| Flag 3 | FLAG3_4f2249c1398d4064ab56e39b0e9b7073 |
| Flag 4 | FLAG4_8ba936c078234a9db2874f5ed241b14f |

### *Flag 1 – HTTP Header Enumeration*

The first flag was discovered by inspecting HTTP response headers using curl. The server revealed identifying information inside the response which contained the hidden flag.

```
└─# curl -I http://target.ine.local
HTTP/1.1 200 OK
Server: Werkzeug/3.0.6 Python/3.10.12
Date: Thu, 12 Mar 2026 20:34:46 GMT
Content-Type: text/html; charset=utf-8
Content-Length: 2557
Server: FLAG1_afa965c7836048cf955a2181f0400af2
Connection: close
```

### *Flag 2 – Hidden Directory Discovery*

The second flag was discovered by identifying a hidden directory exposed by the web server. Accessing the directory revealed a file containing the flag.

```
┌──(root㉿INE)-[~]
└─# curl http://target.ine.local/secret-info/
["flag.txt"]

┌──(root㉿INE)-[~]
└─# curl http://target.ine.local/secret-info/flag.txt
FLAG2_6947ca5b44e447f982d12226b459682e
┌──(root㉿INE)-[~]
└─#
```

### *Flag 3 – Anonymous FTP Access*

The third flag was retrieved through anonymous FTP login. After connecting to the FTP service, directory listing revealed multiple files including flag.txt which contained the flag.

```
ftp> cd /
250 Directory successfully changed.
ftp> ls
229 Entering Extended Passive Mode (|||25272|)
150 Here comes the directory listing.
-rw-r--r--    1 0        0              22 Oct 28  2024 creds.txt
-rw-r--r--    1 0        0              39 Mar 12 20:29 flag.txt
226 Directory send OK.
ftp> get flag.txt
local: flag.txt remote: flag.txt
229 Entering Extended Passive Mode (|||46686|)
150 Opening BINARY mode data connection for flag.txt (39 bytes).
100% |***********************************************************************************************
226 Transfer complete.
39 bytes received in 00:00 (225.36 KiB/s)
ftp>
[2]+  Stopped                 ftp target.ine.local

┌──(root㉿INE)-[~]
└─# ls
Desktop  Documents  Downloads  flag.txt  Music  Pictures  Public  Templates  thinclient_drives  Videos

┌──(root㉿INE)-[~]
└─# cat flag.txt
FLAG3_4f2249c1398d4064ab56e39b0e9b7073
```

### *Flag 4 – Database Enumeration*

The final flag was obtained after retrieving credentials from the FTP server and using them to connect to the MySQL database. Enumerating the database revealed the hidden flag.

```
MySQL [(none)]> ls
    -> cd
    -> show databases
    -> ^C
MySQL [(none)]> show databases;
+--------------------------------------+
| Database                             |
+--------------------------------------+
| FLAG4_8ba936c078234a9db2874f5ed241b14f |
| information_schema                   |
| mysql                                |
| performance_schema                   |
| sys                                  |
+--------------------------------------+
5 rows in set (0.001 sec)
```

## Conclusion

This lab demonstrates how reconnaissance and service enumeration can quickly lead to sensitive information disclosure. Misconfigured services such as exposed headers, accessible directories,

anonymous FTP access, and poorly protected database credentials can significantly increase the attack surface of a system.